
Evaluating the Role of Land Surface Temperature and Other Key Variables in Post-Fire Vegetation Recovery: A Case Study of Mt. Kenya

by

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DECLARATION

I declare that this project is my own work and has not been submitted by anybody else in any other university for the award of any degree to the best of my knowledge.

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Abstract

This study investigated the impact of Land Surface Temperature (LST) and other environmental variables on post-fire vegetation recovery (PVR) in the Mount Kenya region, addressing a significant gap in existing research. With increasing wildfire frequency and severity attributed to climate change and human activities, understanding the dynamics of PVR is critical for ecosystem restoration and management. Utilizing advanced remote sensing techniques, this research developed a predictive model that integrates LST with traditional vegetation indices and other relevant factors such as topography and soil total organic carbon.

Landsat satellite imagery from 2011 to 2021 was used to classify burn severity into seven categories based on the Differenced Normalized Burn Ratio (dNBR) index. Climate data from WorldClimate.org, soil total organic carbon data from Africa Soil Information Service (AFSIS), and topographic variables (elevation, slope, and aspect) were integrated into a Random Forest regression model to predict post-fire recovery trends. The model predicted vegetation recovery using the Normalized Burn Ratio (NBR) for 2011 as the initial condition and the NBR for 2021 as the recovery outcome. Results demonstrated a strong relationship between LST, topographic variables, and vegetation recovery. The Random Forest model achieved an R^2 of 0.9013 and a Root Mean Square Error (RMSE) of 0.0280 on the test dataset, while K-fold cross-validation (10 folds) yielded an average R^2 of 0.8753 and an average RMSE of 0.0406.

Analysis revealed that lower burn severity classes showed faster recovery, with LST emerging as a critical predictor across all severity levels. The combined influence of topographic variables further highlighted the heterogeneous nature of recovery processes, highlighting the need for spatially targeted management strategies. The findings underscore the necessity of incorporating LST into PVR models to capture the complexities of recovery processes in tropical montane ecosystems. This integrated approach informs effective forest management strategies. It further contributes to broader climate adaptation efforts, ensuring the sustainability of Mount Kenya's ecological functions and supporting local communities reliant on its resources.

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Acronyms & Abbreviations

1. AFSIS - Africa Soil Information Service
2. ANOVA - Analysis of Variance
3. BA – Burnt Area
4. CV - Cross Validation
5. dNBR - Differenced Normalized Burn Ratio
6. ETM - Enhanced Thematic Mapper
7. HSD - Honestly Significant Difference
8. KFS - Kenya Forest Service
9. KWS - Kenya Wildlife Service
10. LST - Land Surface Temperature
11. MODIS – Moderate Resolution Spectroradiometer
12. NBR - Normalized Burn Ratio
13. NIR - Near Infrared
14. OLI - Operational Land Imager
15. PVR - Post-fire Vegetation Recovery
16. QGIS - Quantum GIS
17. RF - Random Forest
18. SWIR - Shortwave Infrared
19. SRTM - Shuttle Radar Topography Mission
20. TOTC - Total Organic Carbon
21. TIRS - Thermal Infrared Sensor
22. UNESCO - United Nations Educational, Scientific and Cultural Organization

CHAPTER 1: INTRODUCTION

1.1 Background

Wildfires are a global phenomenon, as evidenced by satellite data sourced from NASA's Fire Information for Resource Management System (FIRMS), illustrating fire locations worldwide. Figure 1 depicts fire events recorded between 27th May and 2nd June 2020, showcasing the spatial distribution of active fires during this period. These wildfires significantly disrupt terrestrial ecosystems, altering vegetation cover, soil properties, and hydrological cycles. High-severity fires often result in more pronounced shifts in forest structure compared to low-severity fires (Barlow and Peres, 2004; Oliveras et al., 2014). These disturbances often result in long-term consequences for biodiversity and ecosystem services, particularly in regions where natural resources are vital for human livelihoods. Communities located in fire-prone areas often face the threat of displacement due to evacuations and property loss during wildfire events (Zhao et al., 2019). Wildfires significantly affect tropical montane forests. Despite disturbances, many dominant species can persist or quickly recolonize post-fire, indicating a degree of ecological resilience (Ford and Hardesty, 2012; Mondal and Sukumar, 2015). Climate change is expected to increase the frequency and intensity of wildfires globally, which may outpace the recovery capabilities of many ecosystems (Chazdon et al., 2016).

In Kenya, Mount Kenya stands out as a critical water tower supplying approximately 40% of the country's water while supporting agriculture, hydroelectric power production, and biodiversity conservation (Henry et al., 2019). The region has experienced several large fires, notably in 2012 (over 100 km²) (Downing et al., 2017). Fires in Mount Kenya have been exacerbated by human activities such as illegal logging, agricultural encroachment, and poaching, which have historically strained the mountain's ecological balance (Henry et al., 2019). Wildfires can lead to significant ecological changes, including alterations in species composition, loss of biodiversity, and changes in soil properties. Beyond the natural environment, wildfires have a substantial influence on people and the local economy (Paveglio et al., 2015).

Post-fire vegetation recovery (PVR) is essential for restoring ecosystem functions and maintaining ecosystem services in regions such as Mount Kenya. Remote sensing has significantly advanced PVR monitoring, providing efficient and timely evaluations across extensive and inaccessible areas. Satellite-based indices, including the Normalized Burn Ratio (NBR) and its derivative, the difference Normalized Burn Ratio (dNBR), are commonly employed to

evaluate fire severity and monitor vegetation recovery. (Pérez-Cabello et al., 2021). These indices leverage changes in spectral reflectance to quantify vegetation damage and regrowth after a fire event, enabling a detailed analysis of ecosystem resilience (Henry et al., 2019). Additionally, soil factors such as changes in soil nutrient levels and structure can affect plant growth and the overall recovery trajectory (Zhao et al., 2023).

The dynamics of PVR are influenced by multiple factors, including fire severity, topography, soil moisture, and climatic variables such as temperature and precipitation. Among these, Land Surface Temperature (LST) is particularly critical due to its role in driving evapotranspiration rates, soil moisture dynamics, and vegetation regrowth (Pérez-Cabello et al., 2021). However, despite its importance, the role of LST in PVR remains underexplored, particularly in tropical montane ecosystems like Mount Kenya. This knowledge gap limits our ability to predict recovery outcomes effectively and develop targeted management strategies.

Mount Kenya's ecological complexity adds another layer of challenge to monitoring PVR. The mountain's vegetation zones, which include lower montane forests, bamboo, upper montane forests, Ericaceous bushlands, and alpine vegetation, respond differently to fire disturbances. For instance, lower montane forests show higher resilience compared to the fire-intolerant bamboo and upper montane forests (Henry et al., 2019). These variations necessitate a tailored approach that accounts for the unique recovery patterns across vegetation types.

Furthermore, the challenges of monitoring PVR in regions like Mount Kenya are compounded by the limited resources available for fire management. Local agencies such as the Kenya Wildlife Service (KWS) and Kenya Forest Service (KFS) often lack spatially explicit fire data, which hinders effective monitoring and decision-making. Remote sensing provides a viable solution by offering detailed and continuous coverage, making it possible to reconstruct fire histories and monitor recovery dynamics over time (Henry et al., 2019).

Given the above challenges, integrating LST and other critical variables into PVR models represents a significant step forward. Such integration can provide deeper insights into the interplay between climatic, topographic, and ecological factors that shape recovery dynamics, ultimately informing more effective forest management and restoration efforts.

Advances in remote sensing technology have improved the ability to monitor wildfires and assess their impacts on ecosystems. Satellite imagery and aerial surveys can provide valuable data on burn severity, recovery rates, and

changes in land cover, aiding in the development of effective management strategies (Dempewolf et al., 2007).

The integration of machine learning algorithms with remote sensing data is becoming increasingly common, allowing for more accurate predictions of fire behavior and post-fire recovery dynamics (Lu et al., 2008).

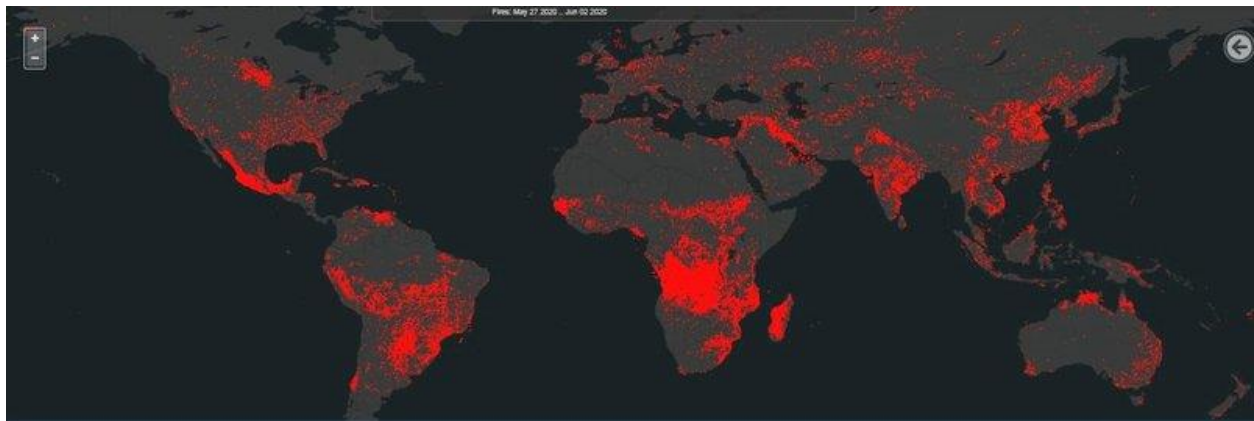


Figure 1: Map of the world showing fire locations recorded by satellite in the 7 days between 27th May and 2nd June 2020. Source NASA firms (Fire Information for Resource Management System).

1.2 Motivation and Problem Statement

Wildfires on Mount Kenya have become more frequent and severe, driven by climate change and anthropogenic pressures. These fires significantly disrupt ecological and hydrological systems, threatening biodiversity and the livelihoods of communities reliant on the mountain's resources (Henry et al., 2019). The March 2012 fire, which is the focus of this study and is shown in Figure 2, destroyed over 8,000 ha of the region. Another fire destruction also occurred around February 2019, as shown in Figure 3 whereby over 80,000 ha was affected. While extensive research has focused on using vegetation indices to monitor PVR, there remains a critical gap in understanding the role of Land Surface Temperature (LST) in this process. LST influences key recovery determinants such as soil moisture and vegetation regrowth but is often overlooked in traditional recovery models (Pérez-Cabello et al., 2021).

Existing models primarily emphasize vegetation indices, neglecting other essential variables like LST, topography, and precipitation. This omission limits the models' ability to capture the complexities of recovery in

heterogeneous ecosystems like Mount Kenya. Moreover, cloud cover and the lack of high-resolution datasets pose additional challenges to monitoring PVR in tropical montane environments (Pérez-Cabello et al., 2021).

To address these gaps, this study proposes the development of a comprehensive PVR model that integrates LST and other key variables. By leveraging satellite imagery and advanced regression models, the research aims to enhance our understanding of PVR dynamics and provide actionable insights for managing post-fire recovery in Mount Kenya. This model will not only fill critical knowledge gaps but also contribute to broader efforts in climate change adaptation and ecosystem restoration.



Figure 2: Mt Kenya Forest Fire in March 2012



Figure 3: Mt Kenya Forest Fire in February, 2019

1.3 Justification

Mount Kenya is a UNESCO World Heritage Site and one of Kenya's most important ecological assets. Its forests regulate water flow, support biodiversity, and contribute to climate regulation through carbon sequestration. However, the increasing frequency and severity of wildfires pose significant threats to these functions, with far-reaching implications for water availability, agriculture, and energy production (Henry et al., 2019). Understanding and enhancing post-fire recovery is, therefore, vital for ensuring the mountain's ecological and socio-economic sustainability.

This study's focus on LST is particularly relevant in the context of climate change. Rising temperatures and altered precipitation patterns are expected to exacerbate fire impacts and complicate recovery processes. By integrating LST into PVR models, this research provides a novel approach to understanding recovery dynamics and addressing the challenges posed by climate variability (Pérez-Cabello et al., 2021).

Moreover, the study aligns with national and global priorities for conserving Kenya's water towers and mitigating climate change impacts. Its findings will offer valuable insights for policymakers and conservation practitioners,

contributing to more effective forest management and restoration strategies. Additionally, the methodologies and lessons derived from this research will have broader applicability, informing PVR studies in other tropical montane ecosystems worldwide.

1.4 Research Identification

1.4.1 General Objective

To analyze the role of Land Surface Temperature (LST) and other environmental variables in post-fire vegetation recovery and develop predictive metrics for monitoring and assessing recovery trends in the Mt. Kenya Forest Region.

1.4.2 Specific Objectives

1. To classify and map the study area into different burn severity classes
2. To evaluate the relationship between LST and other variables in post-fire vegetation recovery using the Random Forest machine learning model.
3. To develop predictive metrics for assessing vegetation recovery potential based on the key environmental variables.

1.5 Research Questions

1. What is the spatial distribution of different burn severity classes in Mt Kenya region?
2. How do LST, burn severity, and other environmental factors affect post-fire vegetation recovery in the Mt. Kenya forest?
3. How accurate is the Random Forest Regression in predicting post-fire vegetation recovery with these variables?

1.6 Study Outline

This study is organized into several chapters. Chapter 2 presents a comprehensive literature review, focusing on previous studies and methodologies relevant to post-fire vegetation recovery. Chapter 3 details the methodology used in this research, including data collection, processing, and analysis techniques. Chapter 4 presents the results and discusses their implications. Finally, Chapter 5 concludes the study, offering recommendations for future research and forest management practices.



CHAPTER 2: LITERATURE REVIEW

2.1 Ecological Impacts of Wildfire and Subsequent Recovery

Wildfires pose a significant threat to ecological security, leading to extensive damage to forest ecosystems. The immediate effects of wildfires include the destruction of vegetation, which can result in habitat loss for numerous species, thereby threatening biodiversity (Zhao et al., 2019). The alteration of habitats can disrupt food chains and ecological interactions, leading to long-term ecological imbalances. Furthermore, wildfires can affect soil quality and water retention, which are critical for sustaining plant life and maintaining ecosystem services (Flannigan & Harrington, 1988). The loss of these services can have cascading effects on local wildlife and plant communities, ultimately impacting the resilience of ecosystems to future disturbances (Bradstock et al., 2009).

The impact of wildfires extends beyond ecological concerns, significantly affecting social development. Communities located in fire-prone areas often face the threat of displacement due to evacuations and property loss during wildfire events (Zhao et al., 2019). The economic implications are profound, as wildfires can lead to substantial financial losses for individuals, businesses, and local governments. Recovery efforts can strain public resources, diverting funds from other essential services (Xu & Ma, 2012). Additionally, the psychological toll on affected populations can be severe, with increased incidences of anxiety and depression reported in communities that have experienced devastating wildfires (Wilhite, 2000). The interplay between ecological damage and social disruption underscores the need for integrated fire management strategies that consider both environmental and human factors.

Meteorological conditions play a critical role in the occurrence and severity of wildfires. Prolonged periods of drought, coupled with high temperatures, create an environment conducive to fire ignition and spread (Zhao et al., 2019). There is a correlation between extreme weather patterns and the incidence of wildfires, hence the need for continuous monitoring of climatic variables (Xu & Ma, 2012). Understanding these relationships is essential for developing effective fire prevention strategies and for informing land management practices that mitigate fire risks (Wilhite, 2000). As climate change continues to influence weather patterns, the frequency and intensity of wildfires are expected to increase, necessitating adaptive management approaches (Trenberth et al., 2014).

Analyzing the severity of wildfires is crucial for understanding their impact on ecosystems and communities (Zhao et al., 2019). Factors such as elevation,

slope, and aspect can affect wind patterns, moisture levels, and vegetation types, all of which contribute to fire dynamics (Zhao et al., 2019). Higher elevation areas are more susceptible to extensive burning, indicating that topographic features play a critical role in fire spread and intensity (Bradstock et al., 2009). Understanding these relationships is essential for land-use planning and fire management, as it allows for the identification of high-risk areas and the implementation of preventive measures (Xu & Ma, 2012). By considering topographic influences, fire management strategies can be tailored to the specific characteristics of the landscape, enhancing their effectiveness in mitigating wildfire impacts.

The advent of remote sensing technology has revolutionized the monitoring and management of wildfires. High-resolution satellite imagery allows for the rapid assessment of burned areas, enabling timely responses to wildfire incidents (Zhao et al., 2019). This capability is crucial for effective disaster management, as it facilitates the identification of high-risk areas and the allocation of resources for firefighting efforts. This information is then vital for post-fire recovery efforts, as it helps prioritize areas for restoration and informs land management decisions aimed at reducing future fire risks (Xu & Ma, 2012). Moreover, remote sensing data can be utilized to assess the extent of damage post-fire, informing recovery strategies and ecological restoration efforts (Flannigan & Harrington, 1988). The recovery of forests post-wildfire is a complex process influenced by various factors, including the severity of the fire and the restoration methods employed. Different burning severities can lead to varying ecological outcomes, with some areas recovering more effectively than others (Zhao et al., 2023). The integration of remote sensing with ground-based observations enhances the accuracy of fire assessments and supports the development of predictive models for future wildfire occurrences (Bradstock et al., 2009).

2.2 The Role of Land Surface Temperature (LST) in Ecosystems and Post-Fire Environments

Land Surface Temperature (LST) refers to the temperature of the Earth's surface as measured from satellite observations. It is a critical parameter in various environmental and climatic studies. LST is typically measured using thermal infrared remote sensing, which allows for the collection of temperature data over large geographic areas. This method is more efficient and cost-effective compared to traditional ground-based measurements (Alqahtani et al., 2023). LST is widely used in remote sensing for environmental monitoring, ecosystem modeling, hydrology, climate studies, weather forecasting, and ecology. It provides continuous and homogeneous data over large areas, crucial for studying surface systems and processes at regional and global scales (Li et al., 2013a; Tang & Li, 2014). LST also affects

the exchange of energy between the land surface and the atmosphere, influencing processes such as evapotranspiration, soil heat flux, and sensible heat flux (Baldocchi et al., 2004; Howell et al., 2010; Kustas & Anderson, 2009; Jia et al., 2001) and subsequently affecting water availability in the soil and atmosphere (Choudhury et al., 1994; Kalma et al., 2008; Ambast et al., 2008)

Additionally, LST is related to vegetation health and stress, with higher values indicating water stress in plants (Zarco-Tejada et al., 2003), and it also affects soil temperature, influencing soil microbial activity, nutrient cycling, and root development (Mexal & South, 1991; Spanos et al., 2000; Kustas & Anderson, 2009).

Changes in LST patterns can indicate climate change, helping to understand the impacts of global warming (Neteler, 2010; Westermann et al., 2012) since LST helps regulate microclimates, with higher vegetation cover typically resulting in lower LST and cooler microclimates (Devine & Harrington, 2007; Marañón-Jiménez et al., 2013b; Gamon et al., 2004). Considering all these factors seems to be a promising indicator for assessing burn severity and monitor post-fire vegetation recovery (De Santis & Chuvieco, 2007; Vlassova et al., 2014; Chuvieco et al., 2006; García et al., 2008).

2.3 Other Key Variables Influencing Post-Fire Recovery

Fire Severity

The intensity and extent of the fire significantly affect recovery. High-severity fires can lead to complete vegetation loss, altering soil properties and microclimates, which may hinder recovery (Pérez-Cabello et al., 2021).

Soil Characteristics

Soil type, texture, and nutrient content play crucial roles in recovery. Fires can change soil chemistry, affecting its fertility and water retention capacity, which in turn influences plant regrowth (Fernández-Guisuraga et al., 2019). Changes in soil nutrient levels and structure can affect plant growth and the overall recovery trajectory (Zhao et al., 2023).

Vegetation Type

Different plant species have varying resilience and recovery strategies. Native species may recover more effectively than invasive species, which can

dominate post-fire landscapes and alter recovery trajectories (Pickell et al., 2016).

Climate Conditions

Temperature, precipitation, and seasonal weather patterns are critical for recovery. Adequate rainfall following a fire can promote rapid regrowth, while drought conditions can delay recovery (Wang & Zhang, 2020). Post-fire climatic conditions, including temperature and precipitation patterns, can significantly impact recovery rates. Favorable weather conditions can enhance regrowth, while adverse conditions may delay recovery (Zhao et al., 2023).

Topography

The landscape's physical features, such as slope, aspect, and elevation, can influence moisture retention and sunlight exposure, affecting how quickly and effectively vegetation recovers (Vogelmann et al., 2012).

Post-Fire Management Practices

Actions taken after a fire, such as reforestation, soil amendments, or erosion control measures, can significantly enhance recovery rates. Conversely, lack of management can lead to prolonged recovery times (Gitas et al., 2012).

Seed Availability and Dispersal

The presence of viable seeds in the soil or nearby areas is essential for recovery. Factors such as animal activity, wind, and water can influence seed dispersal and germination rates post-fire (Dunn et al., 2019).

Phenology

The timing of plant life cycle events (e.g., flowering, leaf-out) can affect recovery. Species that are adapted to respond quickly to post-fire conditions may establish more successfully than those with slower growth rates (Chu & Guo, 2013).

Invasive Species

The introduction or spread of invasive species can alter recovery dynamics. Invasive plants may outcompete native species for resources, leading to changes in community composition and ecosystem function (Banskota et al., 2014).

Microclimate Changes

Fires can create new microclimates by altering light, temperature, and humidity levels in the affected area. These changes can either facilitate or hinder the growth of certain plant species (Tewkesbury et al., 2015).

Soil Properties

The physical and chemical properties of the soil, which can be altered by fire, also influence recovery. Changes in soil nutrient levels and structure can affect plant growth and the overall recovery trajectory (Zhao et al., 2023).

2.4 Remote Sensing and Post-Fire Vegetation Monitoring

Remote sensing technologies (satellites, aerial sensors) provide extensive spatial coverage, enabling monitoring of wildfires across diverse landscapes and ecosystems. This is essential for assessing the extent of fire damage and identifying affected areas (Lee et al., 2024).

Landsat satellites provide a long-term archive of satellite imagery, which is crucial for monitoring changes in vegetation over time (Kennedy et al., 2010). This method allows researchers to analyze the trajectory of vegetation recovery in fire-affected areas, providing insights into the dynamics of forest ecosystems (Pérez-Cabello et al., 2021). The ability to utilize historical data from Landsat enables the assessment of both short-term recovery patterns and long-term ecological trends (Pickell et al., 2016).

Spectral indices, such as the Normalized Difference Vegetation Index (NDVI) and the Normalized Burn Ratio (NBR), are widely used to evaluate vegetation health and recovery post-fire. These indices leverage the reflectance properties of vegetation in different spectral bands, particularly in the red and near-infrared regions, to quantify changes in vegetation cover (Pérez-Cabello et al., 2021). For instance, NDVI is sensitive to chlorophyll content and can indicate the greenness of vegetation, while NBR is effective in assessing burn severity and recovery (Wang & Zhang, 2020). The application of these indices allows for the monitoring of vegetation dynamics and the identification of recovery trends over time (Frazier et al., 2018).

Normalized Difference Vegetation Index (NDVI)

NDVI is one of the most widely used spectral indices for monitoring vegetation health and recovery. It is calculated using the formula: $NDVI = (NIR + Red) / (NIR - Red)$

NDVI values range from -1 to +1, where higher values indicate healthier and denser vegetation. Post-fire, NDVI can help assess the regrowth of vegetation by comparing pre- and post-fire values (Pérez-Cabello et al., 2021).

Normalized Burn Ratio (NBR)

NBR is specifically designed to assess burn severity and monitor recovery in fire-affected areas. It is calculated as: $NBR = (NIR + SWIR) / (NIR - SWIR)$

The index uses the near-infrared (NIR) and shortwave infrared (SWIR) bands, making it sensitive to changes in moisture content and vegetation cover. A decrease in NBR values post-fire indicates burn severity, while increases over time can signify recovery (Wang & Zhang, 2020).

Enhanced Vegetation Index (EVI)

EVI is another index that improves upon NDVI by correcting for atmospheric conditions and soil background effects. It is calculated using: $EVI = G \cdot (NIR - Red) / (NIR + C1 \cdot Red - C2 \cdot Blue + L)$

Here, G is the gain factor, C1 and C2 are coefficients for the aerosol resistance term, and L is the canopy background adjustment. EVI is particularly useful in dense vegetation areas and can effectively track changes in vegetation cover post-fire (Pérez-Cabello et al., 2021).

Half Recovery Time (HRT)

HRT is a post-fire recovery indicator that estimates the time required for vegetation to reach 50% of its pre-fire NDVI levels. This index uses a nonlinear model fitting of post-fire NDVI anomalies to quantify recovery dynamics (Torres et al., 2018). It provides a temporal perspective on recovery, allowing for the assessment of recovery rates across different ecosystems.

Vegetation Recovery Rate (VRR)

VRR is a metric that quantifies the rate of recovery in vegetation cover over time. It is often derived from changes in NDVI or other spectral indices and can help identify areas of rapid recovery versus those that are lagging (Morresi et al., 2019).

NDVI is widely used due to its sensitivity to vegetation greenness and health. It provides a straightforward measure of vegetation cover and is effective in detecting changes over time, making it suitable for monitoring recovery after fire disturbances (Pérez-Cabello et al., 2021).

NBR is specifically designed to assess burn severity and monitor recovery in fire-affected areas. It effectively distinguishes between burned and unburned areas by utilizing the near-infrared and shortwave infrared bands, which are sensitive to moisture content in vegetation. This makes it particularly useful for evaluating the impact of fire and subsequent recovery (Pérez-Cabello et al., 2021).

EVI improves upon NDVI by correcting for atmospheric conditions and soil background effects, making it more reliable in dense vegetation areas. It is sensitive to seasonal variations and can provide a more accurate assessment of vegetation recovery dynamics post-fire (Pérez-Cabello et al., 2021).

FRI2 is a threshold-based index that quantifies forest recovery by analyzing changes in spectral values over time. It is particularly useful for tracking long-term recovery trends and assessing the effectiveness of restoration efforts (Morresi et al., 2019).

VRR quantifies the rate of recovery in vegetation cover, providing insights into how quickly an ecosystem is bouncing back after a disturbance. This metric is valuable for understanding the resilience of different ecosystems and can inform management strategies (Pérez-Cabello et al., 2021).

Remote sensing provides a cost-effective and time-efficient means of collecting data on vegetation recovery compared to traditional field surveys, which can be labor-intensive and limited in scope (Pérez-Cabello et al., 2021). The use of high-resolution satellite imagery enables researchers to capture fine-scale changes in vegetation structure and composition, facilitating a better understanding of post-fire recovery processes (Gitas et al., 2012).

Ground truthing involves collecting field data to compare with remote sensing observations, which helps to confirm the ecological significance of the spectral responses (Bartels et al., 2019). This validation is crucial for establishing reliable connections between remote sensing metrics and actual recovery processes, as different ecosystems may exhibit varying responses to fire disturbances (Szpakowski & Jensen, 2018).

Different ecosystems may respond uniquely to fire, necessitating the use of multiple methods to capture the complexity of recovery dynamics (Pérez-Cabello et al., 2021). For example, integrating spectral data with ecological assessments, such as measuring canopy cover or tree height, can enhance the interpretation of recovery indicators and inform effective management strategies (Pickell et al., 2016).

2.5 Predictive Modeling of Post-Fire Recovery

Predictive modelling aims to forecast future vegetation recovery patterns based on historical data and various influencing factors. This can help in understanding how ecosystems respond to disturbances like wildfires and in planning effective management strategies (Pérez-Cabello et al., 2021). Predictive models often utilize a combination of remote sensing data (e.g., Landsat, MODIS) and ground-based observations. This integration allows for a more comprehensive understanding of vegetation dynamics and recovery processes (Cohen et al., 2018).

Various statistical and machine learning techniques can be employed for predictive modeling, including:

- Regression Analysis: Used to establish relationships between recovery rates and environmental variables (e.g., soil type, moisture levels) (Kennedy et al., 2010).
- Random Forests and Decision Trees: These machine learning methods can handle complex interactions between variables and are effective for classification tasks in ecological studies (Pérez-Cabello et al., 2021).
- Neural Networks: Deep learning approaches can capture non-linear relationships in large datasets, making them suitable for predicting recovery patterns based on spectral indices (Cohen et al., 2018).

Time-series analysis is crucial in predictive modeling for PVR. By analyzing changes over time, models can account for seasonal variations and long-term trends in vegetation recovery (Pérez-Cabello et al., 2021).

Ground-truthing is essential for validating predictive models. This involves comparing model predictions with actual field data to assess accuracy and refine model parameters (Kennedy et al., 2010).

Models should consider various ecological and environmental factors that influence recovery, such as:

- Fire Severity: The intensity and extent of the fire can significantly affect recovery rates (Pérez-Cabello et al., 2021).
- Vegetation Type: Different species have varying resilience and recovery capabilities (Cohen et al., 2018).
- Climate Conditions: Temperature, precipitation, and other climatic factors play a critical role in post-fire recovery (Kennedy et al., 2010).

Predictive models can inform management decisions, such as prioritizing areas for restoration, assessing the effectiveness of post-fire treatments, and understanding the potential impacts of climate change on recovery processes (Pérez-Cabello et al., 2021).

One of the main challenges in predictive modeling for PVR is the inherent variability in ecological responses to disturbances. Models must be flexible enough to accommodate this variability while still providing reliable predictions (Cohen et al., 2018).

2.6 Conceptual Framework and Research Gap

Despite its importance, LST has been underutilized compared to optical data due to challenges in accurate estimation from satellite data, especially in heterogeneous environments (Li et al., 2013a; Tang & Li, 2014). Accurate LST estimation requires precise knowledge of atmospheric conditions and surface emissivity, which is difficult due to variability in land cover types and atmospheric conditions (Jiménez-Muñoz & Sobrino, 2003; Qin et al., 2001). LST's high variability in space and time complicates its accurate characterization using ground measurements alone, and remote sensing still faces challenges in heterogeneous landscapes (Coll et al., 2009; Hook et al., 2004). Differences in observation angles between sensors can lead to discrepancies in LST measurements, affecting data accuracy and its application in post-fire recovery studies (Trigo et al., 2008).

More research is needed to understand the spatial and temporal dynamics of the relationship between burn severity and LST and how this can be used to monitor post-fire recovery (Vlassova et al., 2014; Veraverbeke et al., 2012). Detailed studies are required to quantify the relationship between vegetation regeneration and LST over time and across different ecosystems (Vlassova et al., 2014; Díaz-Delgado et al., 2003). Combining LST with other metrics, such as spectral indices, could improve the assessment of burn severity and post-fire recovery, providing a more comprehensive understanding of fire effects on ecosystems (Chuvieco et al., 2006; García et al., 2008). Comprehensive studies using remote sensing to complement field data over extended periods are needed for both short-term and long-term monitoring of post-fire recovery (Pausas et al., 2009). This study addresses this critical gap by evaluating the combined influence of LST, burn severity, topography, and other environmental factors on post-fire vegetation recovery in the Mt. Kenya forest. This study aims to develop predictive metrics using Random Forest regression, contributing to a more nuanced understanding of post-fire recovery dynamics in this important ecosystem.



CHAPTER 3: MATERIALS AND METHODS

3.1 Study Area

3.1.1 Location and Geographic Context

Mount Kenya, Africa's second-highest peak, is located in central Kenya, approximately 150 kilometres northeast of Nairobi. It is a UNESCO World Heritage Site and a vital ecological region, spanning latitudes 0°07'S to 0°20'S and longitudes 37°03'E to 37°30'E (Nyongesa et al., 2023). Rising to an altitude of 5,199 meters at its highest point, the mountain is characterized by a diverse range of ecosystems that vary with elevation, from tropical forests at the lower levels to alpine meadows and glaciers at the summit. This vertical gradient supports a wide array of flora and fauna, making the region a biodiversity hotspot.

The Mt. Kenya ecosystem is vital for Kenya's socio-economic well-being, serving as a critical water tower for millions of people and key sectors such as agriculture, hydroelectric power, and urban water supply. The mountain is the source of major rivers, including the Tana and Ewaso Nyiro, which sustain the livelihoods of downstream communities. Moreover, the region is rich in volcanic soils, which support productive agricultural practices in the surrounding areas.

From an ecological perspective, Mt. Kenya's forests play a crucial role in carbon sequestration and maintaining regional climate stability (Ndalila et al., 2022). However, the region is under threat from human activities such as deforestation, land-use changes, and encroachment. These activities have heightened the risk of forest fires, with significant implications for biodiversity, soil stability, and water resources.

Climatically, Mt. Kenya experiences a bimodal rainfall pattern, with the long rains occurring between March and May and the short rains from October to December. The region's climate is influenced by its elevation, with temperatures decreasing significantly at higher altitudes (Nyongesa et al., 2023). This climatic variability makes Mt. Kenya an ideal study area for understanding the effects of environmental changes, particularly in the context of post-fire vegetation recovery and land surface dynamics.

The mountain's diverse ecosystems, socio-economic importance and vulnerability to climate change and anthropogenic pressures, make it a compelling site for research. Studies conducted here, such as vegetation recovery following disturbances like fires, provide valuable insights that can inform conservation efforts and sustainable management practices, not only for Mt. Kenya but also for other highland ecosystems in Africa.

3.1.2 Ecological Characteristics

Mount Kenya is a volcanic mountain with distinct vegetation zones shaped by elevation and climatic conditions. Its ecosystem plays a critical role as one of Kenya's vital water towers, providing approximately 40% of the country's water supply and supporting biodiversity, agriculture, and hydroelectric power generation (Henry et al., 2019). Mount Kenya's forests are integral to Kenya's water supply, hydropower production (60% of the national electricity supply), and biodiversity conservation. Fires, primarily human-caused, threaten these services by disrupting hydrological cycles and biodiversity. Past surveys revealed extensive damage to the bamboo and montane forests due to illegal activities and recurring fires (Henry et al., 2019).

Vegetation Zones

The vegetation on Mount Kenya is stratified into distinct zones based on altitude, each characterized by specific flora and ecological conditions:

Lower Montane Forest (1200–2500 m): This zone borders agricultural settlements and resembles tropical rainforests in the Congo Basin (Notter et al., 2007). It is highly biodiverse but has been significantly impacted by human activities such as logging, livestock grazing, and illegal cultivation (Henry et al., 2019).

Bamboo Zone (2200–3200 m): The bamboo forest, dominated by *Arundinaria alpina*, is well-developed and less fragmented compared to other East African mountains. However, it is sensitive to fire and human encroachment (Henry et al., 2019).

Upper Montane Forest (2500–3500 m): These open-canopy forests are dominated by African pencil cedar (*Juniperus procera*) and dense grasses. They are less resilient to frequent fire disturbances (Henry et al., 2019).

Ericaceous Zone (3500–4000 m): Located above the tree line, this zone consists of shrubs adapted to fire, such as *Erica* species, which resprout post-fire, although they require longer recovery times compared to other vegetation types (Niemelä & Pellikka, 2004).

However, excessive fires can lead to grass invasion and ecosystem degradation (Henry et al., 2019).

Alpine and Nival Zones (above 4000 m): The Paramo or alpine zone is characterized by dwarf vegetation, transitioning into the Nival zone of volcanic rocks and glacial landscapes.

3.1.3 Fire History

Fires have been a natural part of the Mount Kenya ecosystem for thousands of years, with paleoecological evidence indicating that they have occurred in the area since at least 26,000 years before the present (BP) day (Nyongesa et al., 2023). However, the frequency and intensity of fires have increased in recent decades, primarily due to human activities (tourism, poaching, and pastoralism) (Downing et al., 2017).

A 1999 aerial survey (conducted from February to June) documented substantial damage to Mount Kenya's montane forests. The assessment showed significant harm from illegal logging, especially affecting bamboo and lower montane areas. It also revealed forest clearing for small-scale farming at the forest's edge and illegal cannabis cultivation within the forest interior (Kenya Wildlife Service, 2012). During this five-month aerial survey, 21 active fires were observed, and fire scars were evident in the drier forest zones on the northern and western slopes of the mountain (Kenya Wildlife Service, 2012).

In recent years, large fires have become increasingly common on Mount Kenya, with significant burned area totals reported:

- In 2005, approximately 3000 hectares of moorland burned.
- In 2009, around 2500 hectares of moorland were affected.
- In 2011, a substantial area of 3600 hectares of upper montane forest was reported to have burned (Henry et al., 2019).
- By March 2012, it was reported that over 8000 hectares had burned following delayed rains, indicating a concerning trend in fire frequency and intensity (Njagi, 2012).
- In March 2019, approximately 80000 hectares.

The Kenya Wildlife Service (KWS) is primarily responsible for fire suppression on Mount Kenya. However, the agency faces significant resource limitations that hinder effective firefighting efforts. Historical fire data may not significantly aid their firefighting strategies, as many areas have burned multiple times within a decade. This indicates that fuel loads are not necessarily limited by recent fires, complicating the management of fire risks in the region (Davies et al., 2009).

3.2 Datasets

DATASET	SOURCE	SPATIAL RESOLUTION	ROLE/ DESCRIPTION	USE/
Landsat 7 ETM+ Landsat 8 OLI/TIRS	United States Geological Survey(USGS)	30m	NBR/dNBR Calculation LST Retrieval	
MCD64A1	National Aeronautics and Space Administration (NASA)	500m	Burnt Area Product	
Shuttle Radar Topography Mission (SRTM)	United States Geological Survey(USGS)	30m	Topographic variables (Elevation, slope, Aspect)	
Climate Data	WorldClimate.org	1Km	Climatic variables(Temperature, Precipitation)	
Soil Data	Africa Soil Information Service (AFSIS)	250m	Soil Total Organic Carbon	
Field Data	Kenya Forest Service (KFS)		validation	

Table 1: Datasets

3.3 Data Description

3.3.1 Landsat 7 & Landsat 8

For this study, Landsat imagery was utilized to derive key indicators such as the Normalized Burn Ratio (NBR), differenced NBR (dNBR), and Land Surface Temperature (LST) to assess burn severity and post-fire vegetation recovery. The data spans the years 2011 to 2021, with Landsat 7 ETM+ providing imagery for the years 2011 and 2012, and Landsat 8 OLI/TIRS used for imagery from 2013 onwards. To calculate NBR, bands 4 (0.77 – 0.90 μm) and

7 (2.09 – 2.35 μm) for Landsat 7 ETM+ and bands 5 (0.85 – 0.88 μm) and 7 (2.11 – 2.29 μm) for Landsat 8 OLI/TIRS (Near-infrared and Shortwave infrared respectively each) were used. Healthy vegetation reflects high levels of NIR and absorbs SWIR due to its moisture content. This results in high positive NBR values. Burned areas exhibit reduced NIR reflectance (due to vegetation loss) and increased SWIR reflectance (due to exposed soil or charred material). This results in low or even negative NBR values. 2011 was selected as the pre-fire year while the rest of the years were post-fire years. The difference between pre-fire and post-fire NBR values (**dNBR**) was then used to classify the severity of burns, according to the United States Geological Survey (USGS) burn severity thresholds.

Class	dNBR Range	Description
Unburned	< -0.1	Areas unaffected by fire.
Low Severity (Regrowth)	-0.1 to 0.1	Low impact or areas experiencing regrowth.
Low Severity (Burned)	0.1 to 0.27	Minor fire effects.
Moderate Severity	0.27 to 0.44	Moderate vegetation and soil burn.
High Severity	0.44 to 0.66	Significant vegetation and soil impact.
Very High Severity	> 0.66	Severe vegetation and soil damage.
Increased Greenness	Negative dNBR values < -0.1	Areas showing increased vegetation (e.g., post-fire recovery or seasonal changes).

Table 2: United States Geological Survey (USGS) Burn Severity Classes

For Land Surface Temperature (LST), band Band 6 (Thermal Infrared): 10.40 – 12.50 μm was used for Landsat 7 ETM+ while bands 10 (Thermal Infrared Sensor 1): 10.60 – 11.19 μm and 11 (Thermal Infrared Sensor 2): 11.50 – 12.51 μm for Landsat 8 OLI/TIRS) were used in the LST calculations. The thermal band data was scaled to temperature (Kelvin), and the mean LST for the entire year is calculated. Finally, the LST is exported as a GeoTIFF image for each year to Google Drive.

3.3.2 Climate Data

Temperature data (Minimum, maximum, and mean temperature data) and Precipitation data were downloaded directly from the worldclim.org website in raster format at a spatial resolution of approximately 1 km.

3.3.3 Soil Data

The Total Organic Carbon (TOTC) dataset provides critical information on the amount of organic carbon present in the soil, which directly impacts vegetation growth and recovery by influencing soil fertility, water retention, and nutrient availability. Understanding the Total Organic Carbon in the context of fire recovery provides insights into carbon sequestration rates, which can help quantify the potential for long-term recovery in terms of carbon storage.

The data was obtained from the Africa Soil Information Service (AFSIS), which provides high-quality soil data at 250m resolution for various regions across Africa.

3.3.4 SRTM

The Shuttle Radar Topography Mission (SRTM) dataset provides high-resolution elevation data, which is critical for understanding terrain characteristics and their influence on ecological processes. For this study, SRTM data was utilized to analyze the topographic variability within the study area, which plays a role in post-fire vegetation recovery. The image data was obtained directly from USGS.

From there, QGIS was used to derive aspect and slope values from the DEM data.

3.3.5 Kenya Forest Service Data

Kenya Forest Service (KFS) provided field data that was used to validate the spatial extent and burn dates of burn areas derived from satellite data

3.3.5 MCD64A1

The MCD64A1 Fire Product detects burned areas using algorithms that analyze the spectral and temporal characteristics of MODIS data. It identifies changes in vegetation and surface properties caused by fire, specifically targeting areas where biomass burning has occurred. The product detects burned areas by analyzing changes in MODIS Surface Reflectance Bands and Thermal Anomalies from the MOD14/MYD14 Active Fire Product. The algorithm compares pre- and post-fire reflectance, identifying burned areas with lower visible and near-infrared reflectance and higher shortwave infrared reflectance. It combines daily MODIS observations to track fire evolution and reduce false positives, assigns a Julian date to burned pixels based on the first spectral change, and assigns confidence levels based on signal strength and algorithm performance.

3.4 Overall Methodology

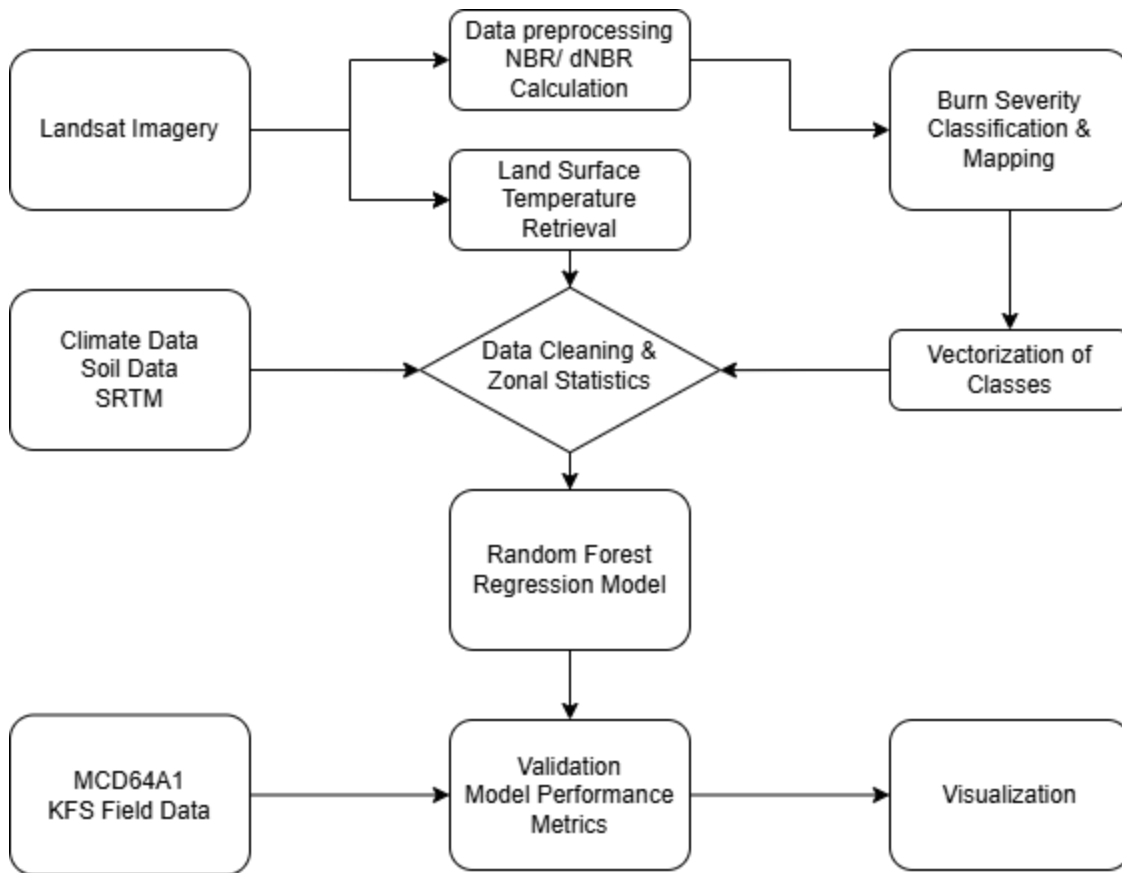


Figure 5: Overall Workflow for Post-Fire Vegetation Recovery Analysis

Abbreviations: NBR-Normalized Burn Ratio; dNBR-Differenced Normalized Burn Ratio; SRTM-Shuttle Radar Topography Mission; KFS-Kenya Forest Service; MCD64A1-Moderate Resolution Imaging Spectroradiometer(MODIS) Burned Area Product

3.5 Methodology Description

3.5.1 Data Acquisition

Landsat imagery for pre and post-fire conditions and SRTM data were downloaded from the USGS Earth Explorer platform.

MCD64A1 product was obtained through the Google Earth Engine Catalogue. Climate data was downloaded from <https://www.worldclim.org> website at a resolution of 2.5 minutes. For the soil data, the total organic carbon data was downloaded from Africa Soil Information Service (AFSIS) at <https://www.isric.org> website at a resolution of 250m. Field data for validation was obtained from KFS which included burnt area extents.

3.5.2 Data Pre-processing

The Landsat Collection 2 Level-2 imagery was processed using Google Earth Engine, where cloud and snow masks were applied, and images from 2011 to 2021 were extracted and clipped to the area of interest. For the climate data, raster calculations were done on the images to obtain the final required variables, which were mean annual maximum temperature, mean annual minimum temperature, mean annual temperature and mean annual precipitation sum

The soil total organic and DEM data were clipped to the area of interest. The DEM image was then used to extract slope and aspect images in QGIS.

3.5.3 Burn Severity Classification and Mapping

3.5.3.1 Index Calculation

Normalized Burnt Ratio (NBR) was calculated in Google Earth Engine using the formula: **$NBR = (NIR - SWIR) / (NIR + SWIR)$** for the prefire and postfire images. These images were exported and input to QGIS where the dNBR was calculated using the following formula: $dNBR = NBR_{prefire} - NBR_{postfire}$.

3.5.3.2 Burn Severity Classification

Thresholds for dNBR values were used to classify the area into seven burn severity classes as recommended by USGS: unburned, low, moderate-low, moderate-high, high, very high, and extreme severity. This was done in the Raster Calculator in QGIS, where the values were reclassified to values 1 to 7 to represent the classes. Subsequently, maps for the post-fire years were created in the layouts.

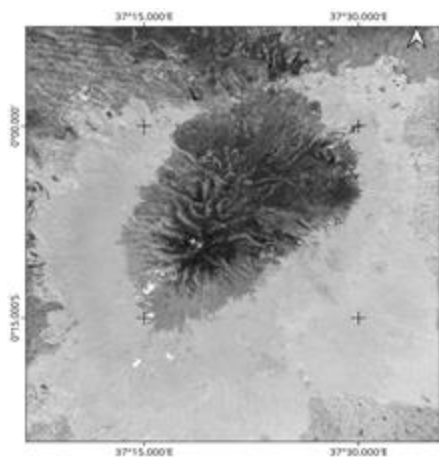


Figure 6: Pre-fire Image, 2011

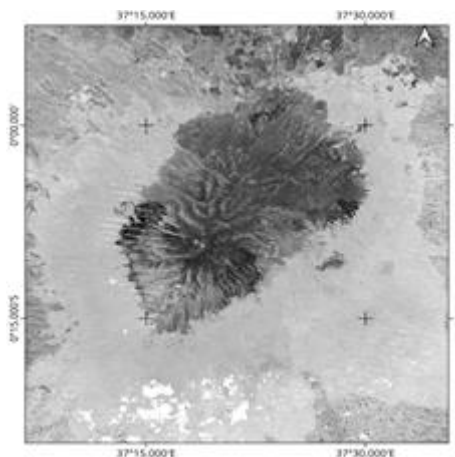


Figure 7: Post-fire Image, 2012

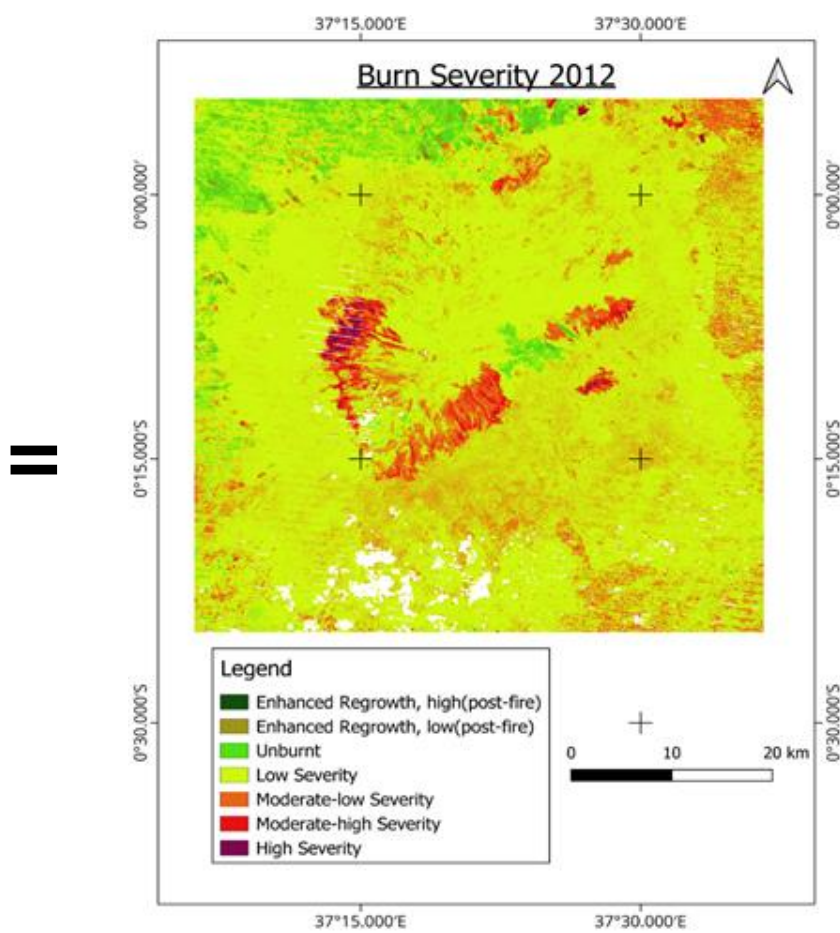


Figure 8: Classified Burn Severity Map, 2012

3.5.4 LST Retrieval

LST was derived from Landsat thermal bands using Google Earth Engine's Surface Temperature algorithm. The values were rescaled and matched to the spatial resolution of the NBR dataset.

3.5.5 Model Development

3.5.5.1 *Overview of the Model*

In this study, a Random Forest Machine learning model was employed to predict post-fire vegetation recovery in the Mt. Kenya region using various environmental predictors. Random Forest is a supervised machine learning algorithm that creates multiple decision trees during training and averages their outputs for robust predictions. It is particularly suited to this study due to its ability to handle high-dimensional data and complex, non-linear relationships between variables.

3.5.5.2 *Application in This Study*

The model was used to predict Normalized Burn Ratio (NBR) values representing vegetation recovery for the years 2012–2021, with 2011 as the pre-fire baseline. The predictors included Land Surface Temperature (LST), dNBR (burn severity), and other environmental variables derived from Landsat imagery and supplementary datasets.

3.5.5.3 *Dataset Preparation*

The dataset used for model training consisted of observations collected from the Mt. Kenya region over a span of years (2011–2021) saved in CSV format. Key steps in data preparation included:

Vectorization

The reclassified raster was converted to polygons using the Raster to Vector tool in QGIS. This vector polygon layer is what was used to get zonal statistics of each dataset for each burn severity class across the years.

Polygon_vectors — Features Total: 93582, Filtered: 93582, Selected: 0

	fid	DN
1	1	2
2	2	3
3	3	2
4	4	2
5	5	3
6	6	2
7	7	3

Figure 9: Screenshot of the Polygon Vectors Attribute Table

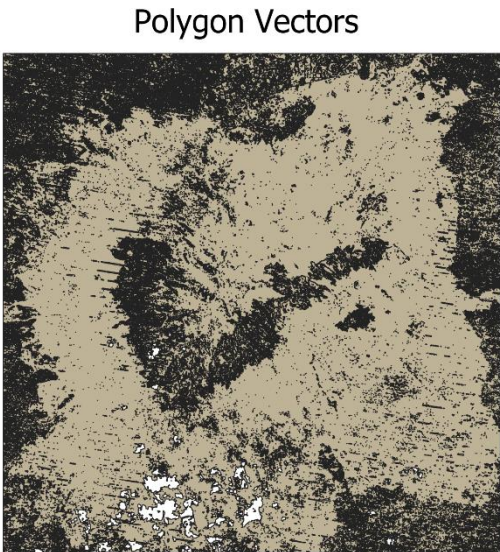


Figure 10: Polygon Vector

Independent Variables (Predictors)

- Land Surface Temperature (LST)
- Burn Severity (dNBR)
- Additional Variables: Maximum temperature, minimum temperature, mean temperature, precipitation sum, soil total organic carbon, elevation, aspect, slope.

Dependent Variable (Target)

- The Normalized Burn Ratio (NBR) was used to measure vegetation recovery, with the NBR value for the final year representing post-recovery.

The predictors and target variables were combined into a structured dataset, with each row representing a unique data point (e.g., zonal average for a burn severity class in a specific year).

3.5.5.4 Model Structure

The Random Forest regression model operates as an ensemble of decision trees. Key structural components include:

Number of Trees (n_estimators)

A total of 100 trees were used in the forest. This parameter controls the size of the ensemble and impacts both accuracy and computational cost. The model used 100 trees so as to balance computational efficiency and predictive accuracy, as a common baseline for capturing complex relationships in the data while avoiding overfitting or underfitting.

Tree Depth (max depth)

The maximum depth of each tree was set to prevent overfitting by limiting how granular the splits could become.

Split Criteria

Each tree splits the dataset recursively based on a feature and a threshold that minimizes prediction error. This process continues until a stopping condition is met (e.g., maximum depth or minimum samples per leaf).

Feature selection at splits

At each node, a random subset of features was considered for splitting, ensuring diversity among the trees and preventing overfitting.

3.5.5.5 Implementation Workflow

Data Splitting:

- The dataset was split into **training** (80%) and **testing** (20%) subsets.
- Training data was used to build the model, while testing data was reserved for evaluating predictive performance.

Bootstrapping:

- The training dataset was resampled using bootstrapping, creating multiple subsets for training individual trees. This step ensured that each tree saw a slightly different view of the data.

Tree Construction:

- For each bootstrapped subset, a decision tree was built by:
 - Selecting random features at each split.
 - Splitting data into smaller subsets based on thresholds that minimized error (mean squared error for regression).

Ensemble Prediction:

- For each input observation, predictions were made by all trees in the forest.
- The final prediction was computed as the average of all individual tree predictions, improving accuracy and stability.

Hyperparameter Tuning:

- **Grid Search:** A systematic search across hyperparameter combinations was conducted to find the best values for `n_estimators` and `max_depth`.

- **Cross-Validation:** K-fold cross-validation was applied to ensure the model's performance generalized well across different subsets of the data.

3.5.5.6 Model Training and Validation

Training the Model

The training subset was fed into the Random Forest regression algorithm, which iteratively built the ensemble of decision trees. The model learned to map the relationships between predictors and the NBR values (target variable).

Testing the model

The testing subset was used to evaluate how well the model generalized to unseen data. Metrics such as RMSE and R^2 quantified the accuracy and explanatory power of the model.

Feature Importance Analysis

The model provided feature importance scores, revealing which predictors (e.g., LST, dNBR) contributed most to explaining post-fire vegetation recovery.

3.5.5.7 Model Structure Representation

Here's a conceptual representation of the Random Forest regression model in this study:

Input Layer

Contains all independent variables:

- LST, dNBR, and other environmental predictors.

Decision Trees ($n_estimators = 100$)

Each tree in the forest:

- Receives a bootstrapped subset of data.
- Evaluates splits based on a subset of features.
- Outputs a predicted NBR value.

Output Layer (Ensemble)

- Combines predictions from all 100 trees using averaging.

- Produces a final predicted NBR value for each input observation.

3.5.5.8 Feature Importance

The model's feature importance scores ranked predictors based on their contribution to explaining vegetation recovery. LST and dNBR were identified as key variables influencing recovery patterns.

3.5.5.9 Model Evaluation

The dataset was split into training (80%) and testing (20%) subsets.

The model's accuracy was evaluated using:

Root Mean Squared Error (RMSE)

R² (Coefficient of Determination)

Cross-validation was also done using the K-fold CV to ensure that these metrics are consistent across multiple data partitions, ruling out the possibility of an overly favourable or biased split.

3.6 Accuracy Assessment

Validation Data: Field data collected by KFS was used to validate the accuracy of the burn severity classification

Accuracy Metrics: The accuracy of the burn severity classification was assessed using a confusion matrix, calculating overall accuracy, producer's accuracy, and user's accuracy. The accuracy of the Random Forest model was assessed using the Root Mean Square Error (RMSE) and R-squared.

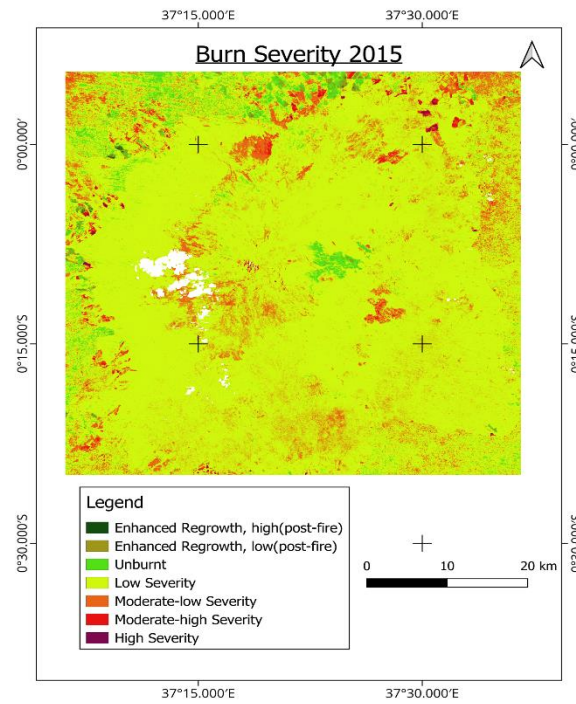
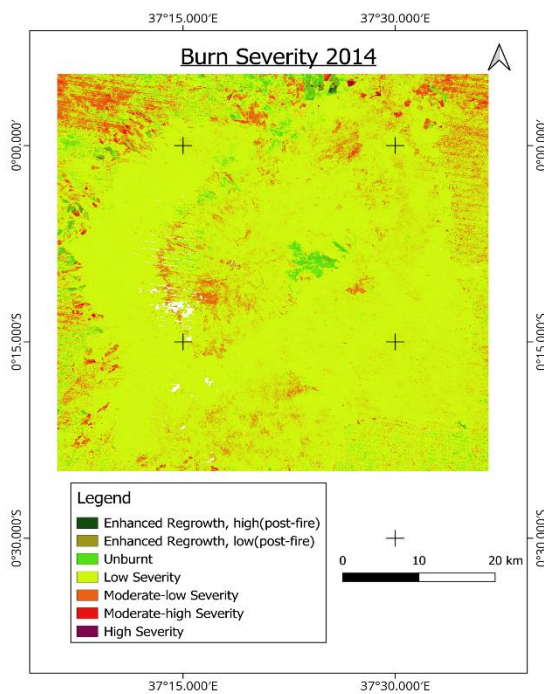
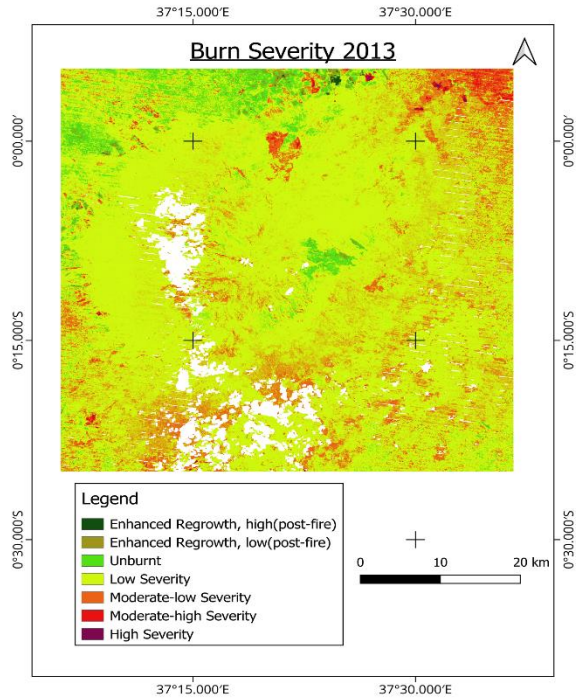
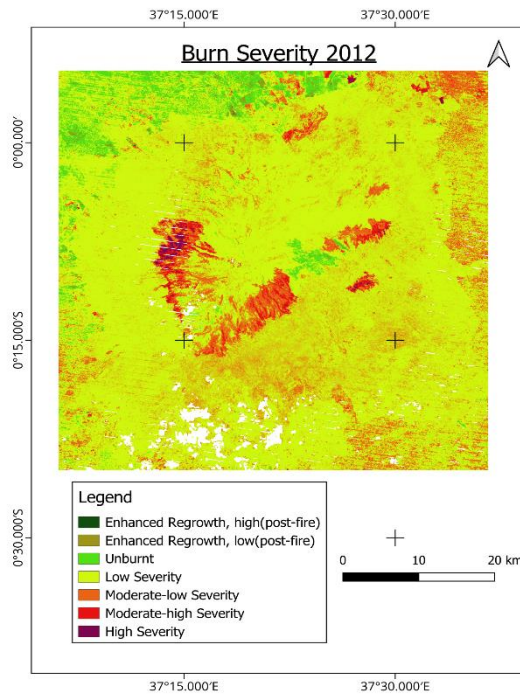
3.7 Software and Tools

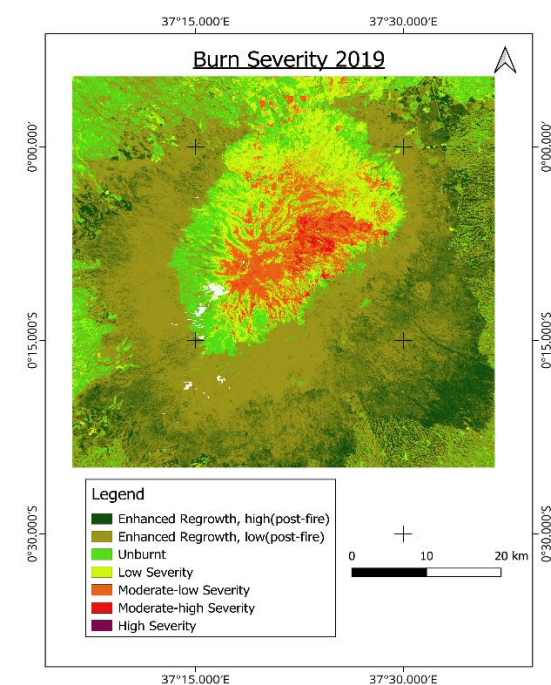
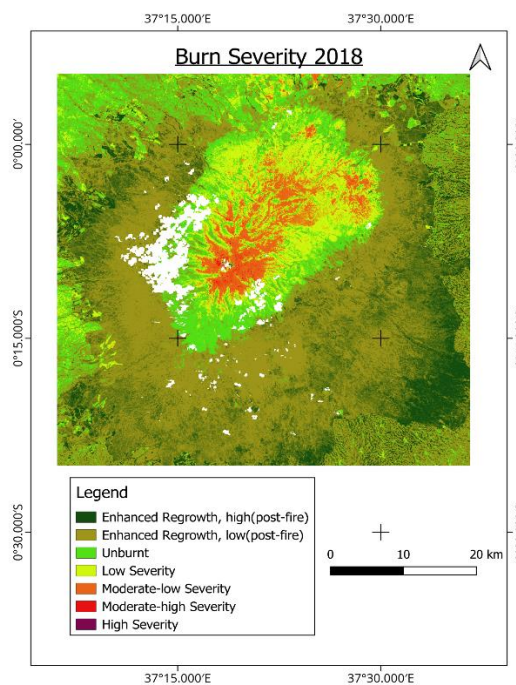
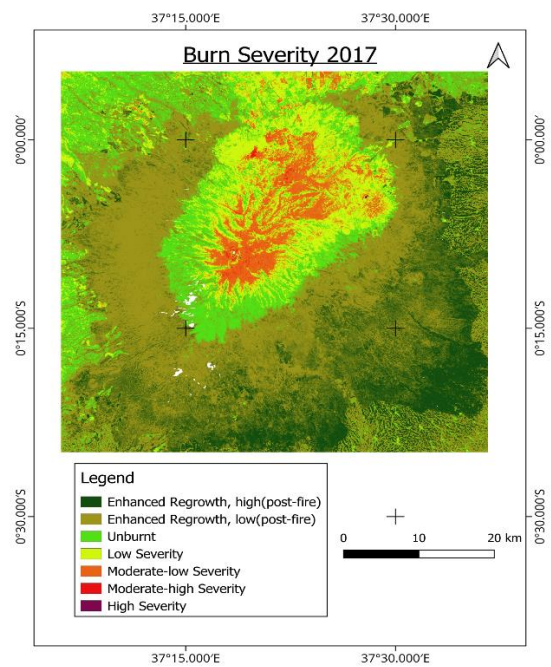
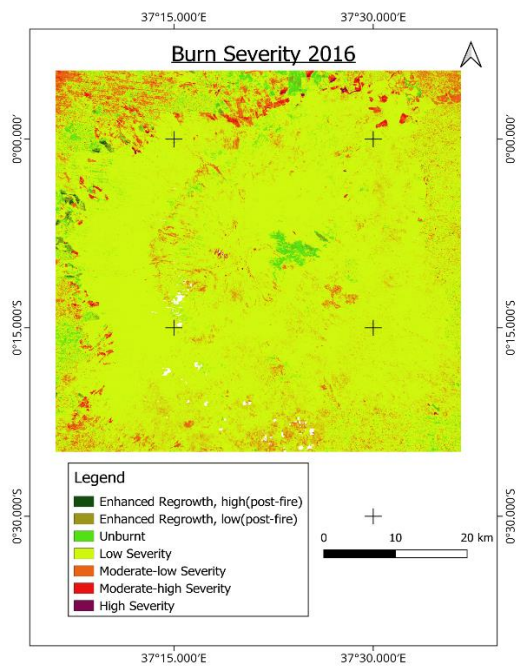
The following software and tools were used for data processing and analysis:

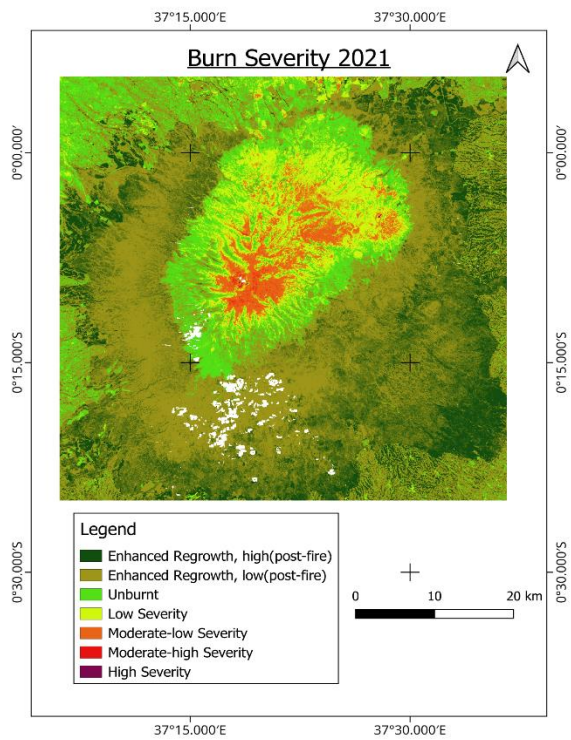
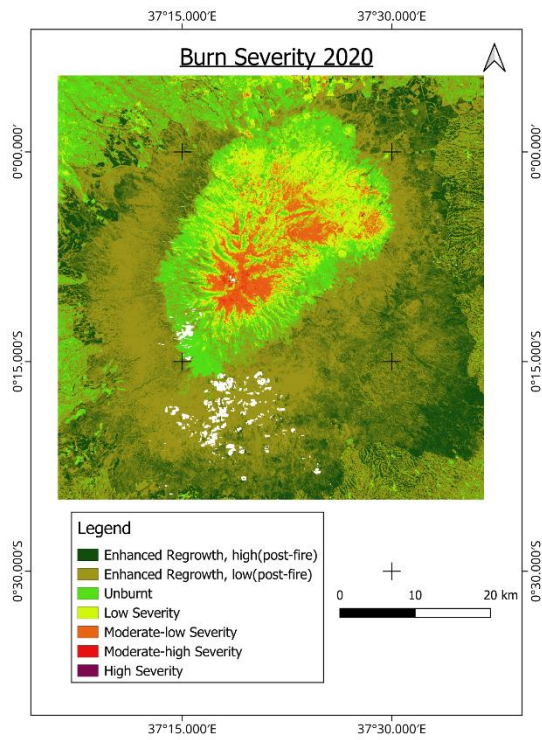
- QGIS
- Python
- Google Earth Engine

CHAPTER 4: RESULTS

4.1 Burn Severity Maps







4.2 Time Series

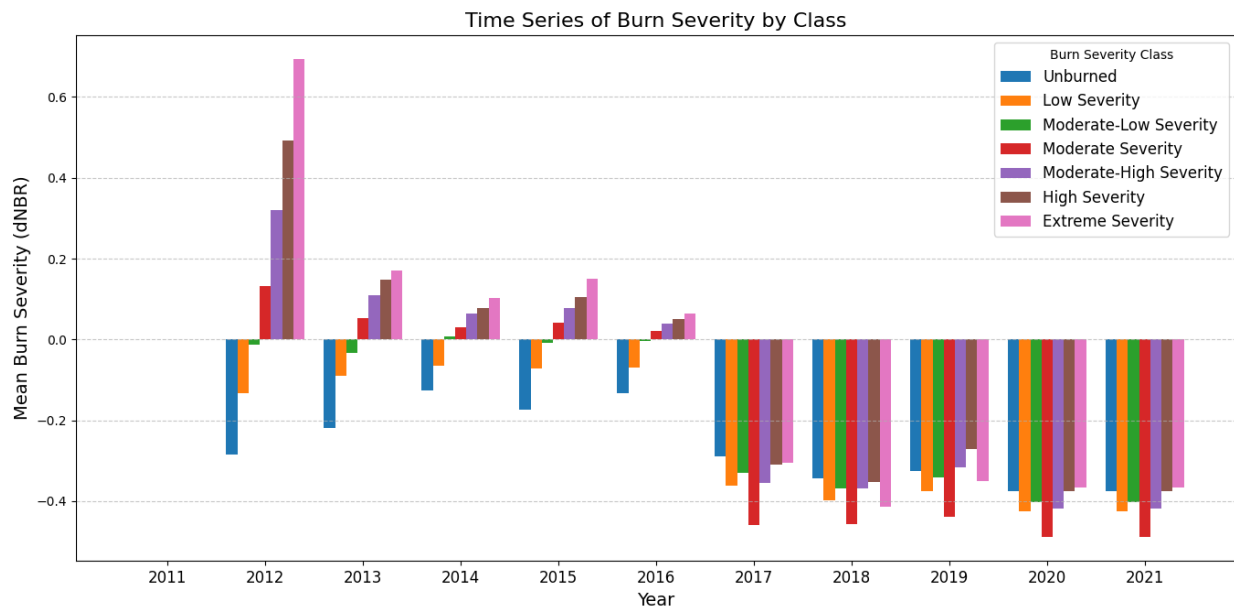


Figure 11: Time Series for Burn Severity

4.3 Variable Importance

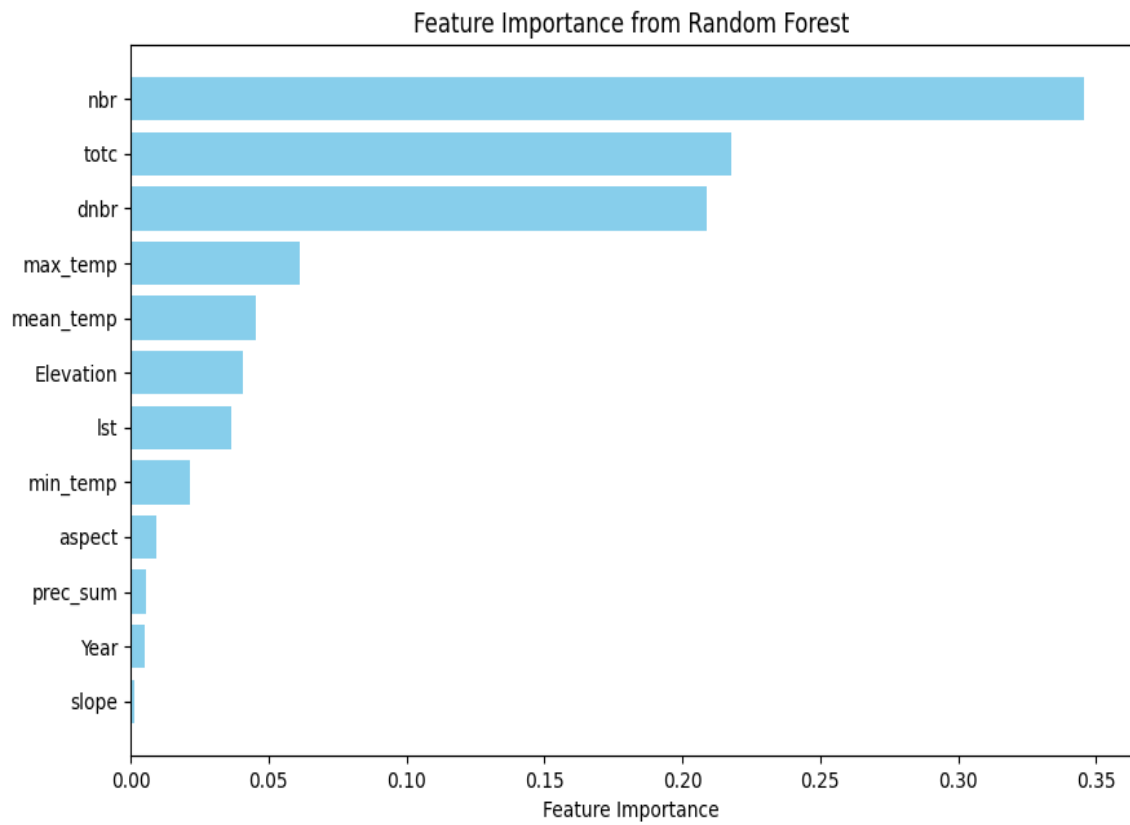


Figure 12: Feature Importance from Random Forest

4.4 Correlation Matrix

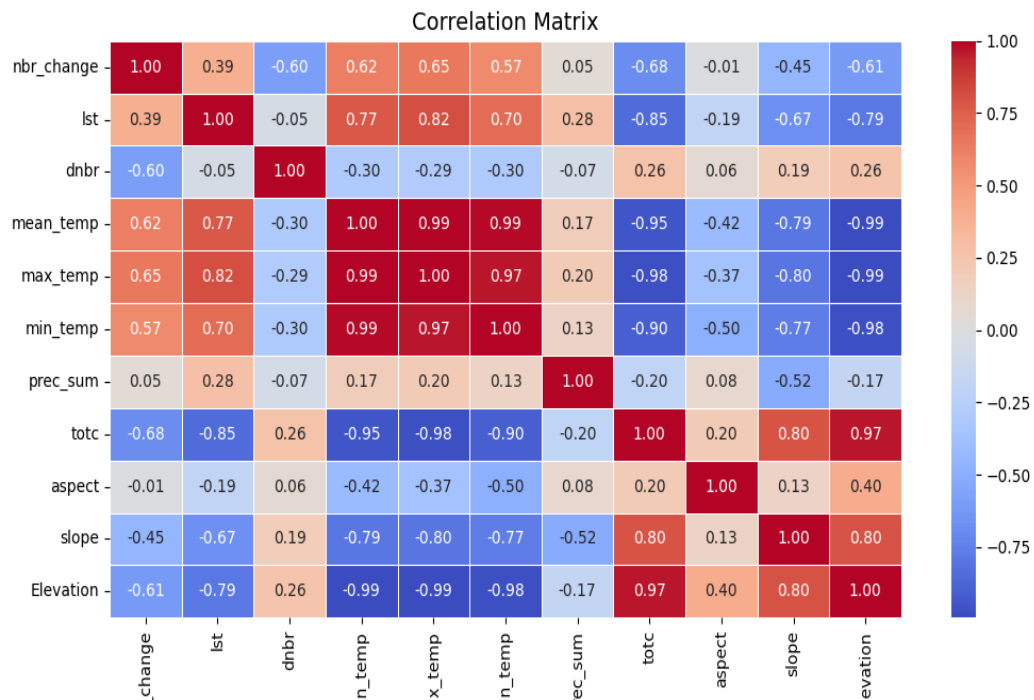


Figure 13: Correlation Matrix of the Variables

4.5 Residual Plot

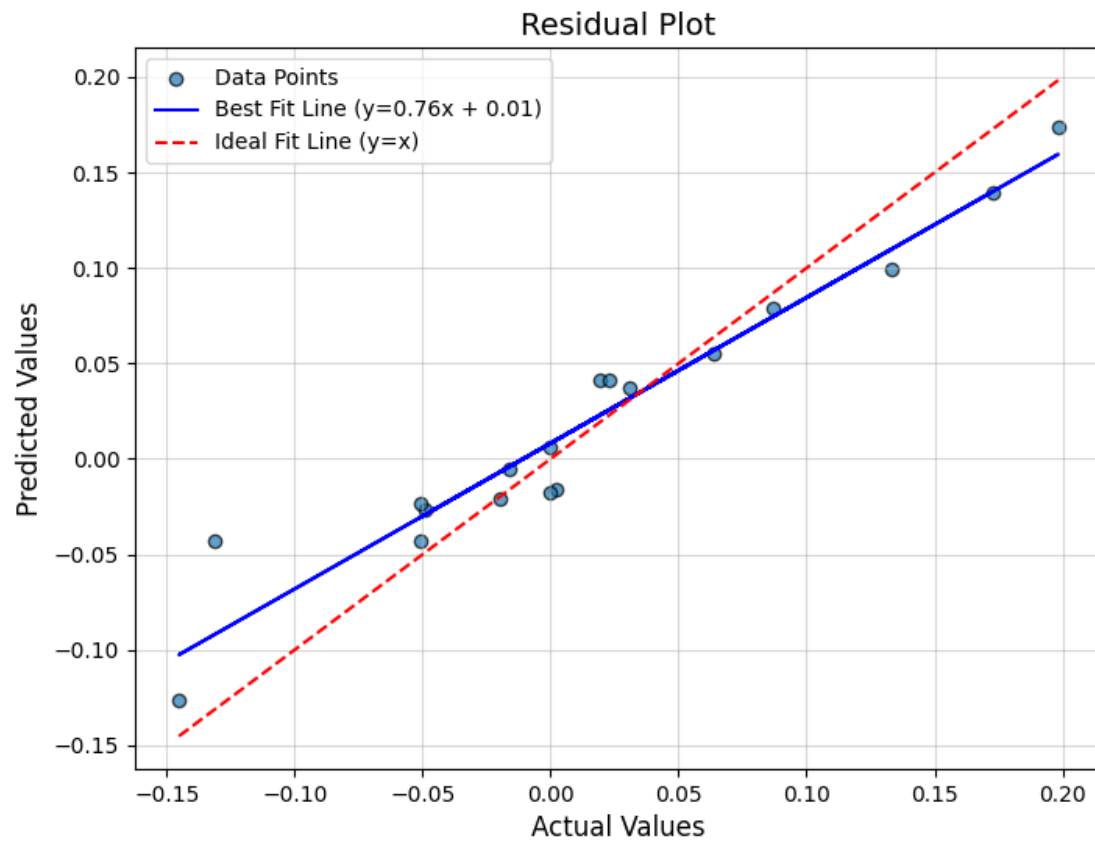


Figure 14: Residual Plot

4.6 Performance Metrics

RMSE: 0.028007673474642072

R^2 : 0.901325203720175

K-Fold Cross-Validation Results (10 Folds):

Average RMSE: 0.0406

Average R^2 : 0.8753

F-statistic: 12.163610573978437, p-value: 1.7563463215895597e-10

Multiple Comparison of Means - Tukey HSD, FWER=0.05

4.7 Tukey HSD Plot

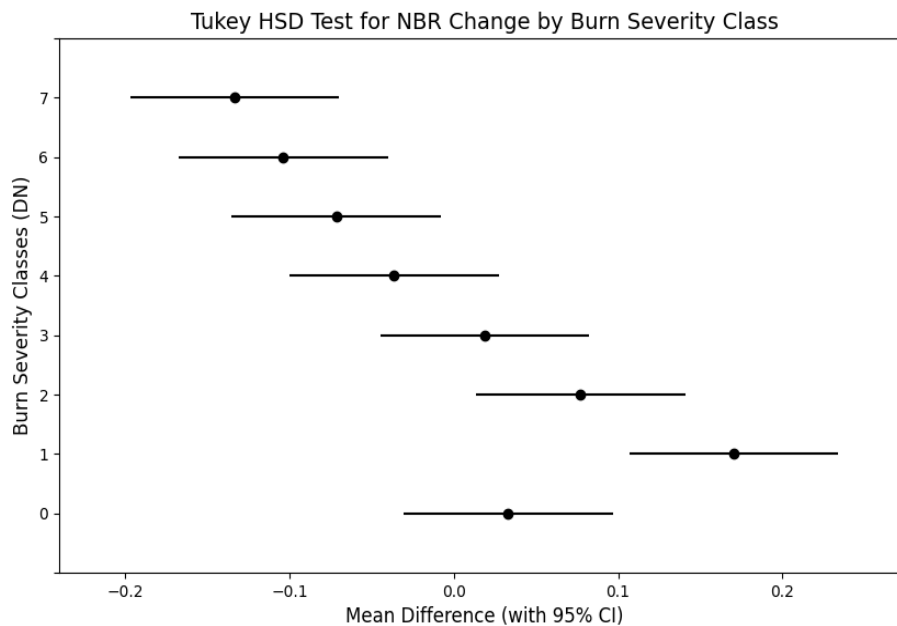


Figure 15: Tukey HSD Plot

4.8 Recovery Plot

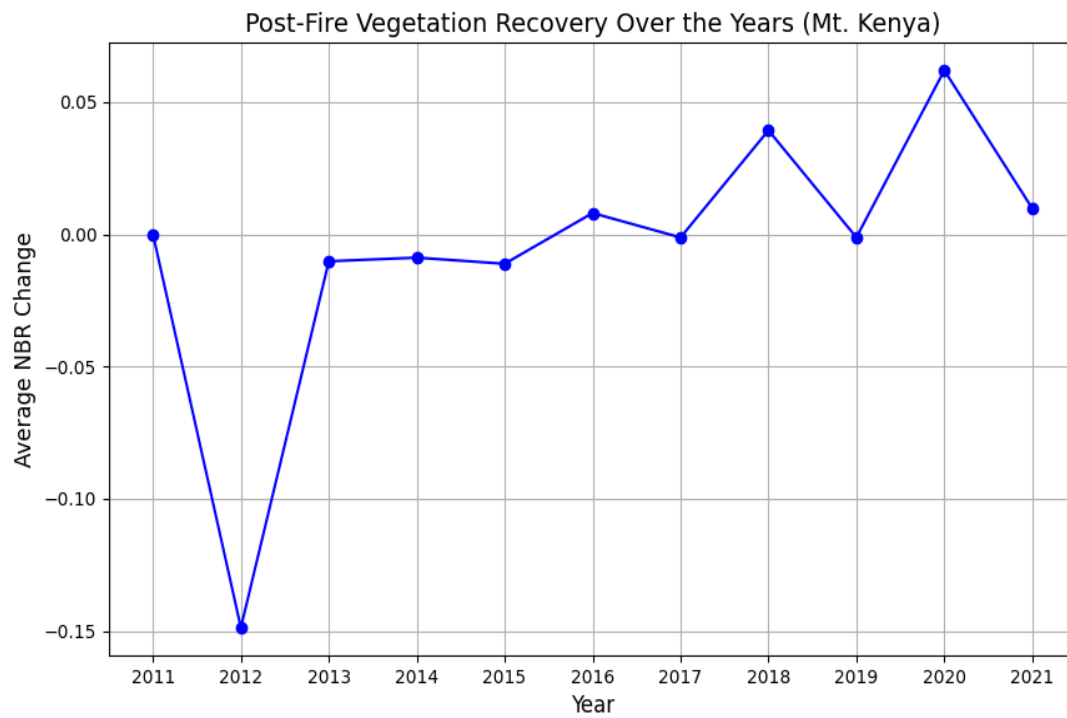


Figure 16: Recovery Plot

4.9 Recovery by Burn Severity Class

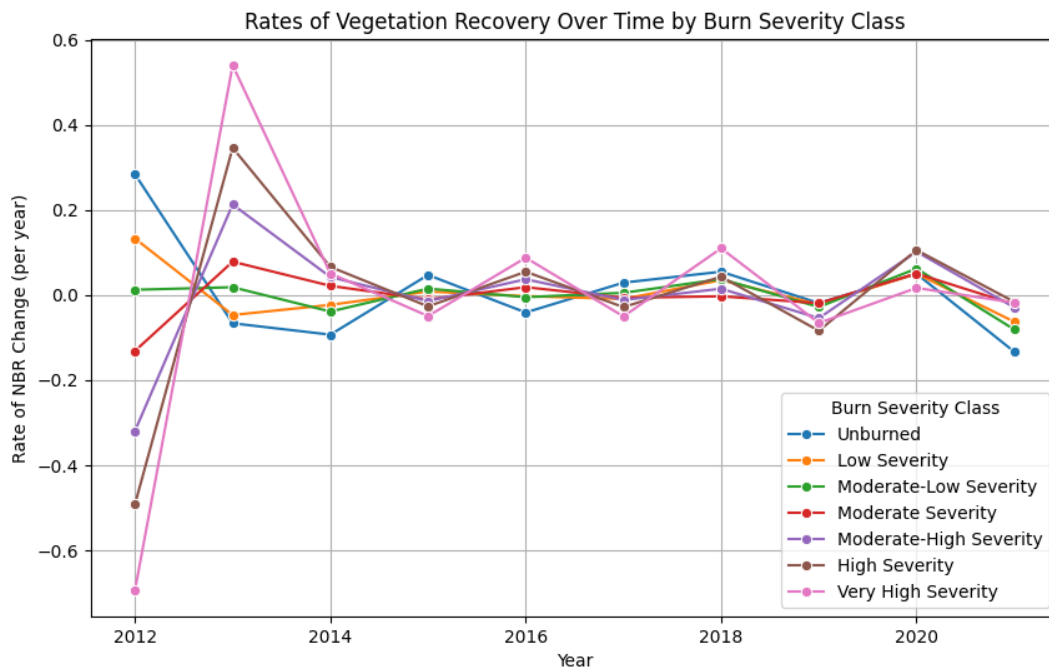


Figure 17: Recovery Plot by Severity Class

4.10 Partial Dependence Plot

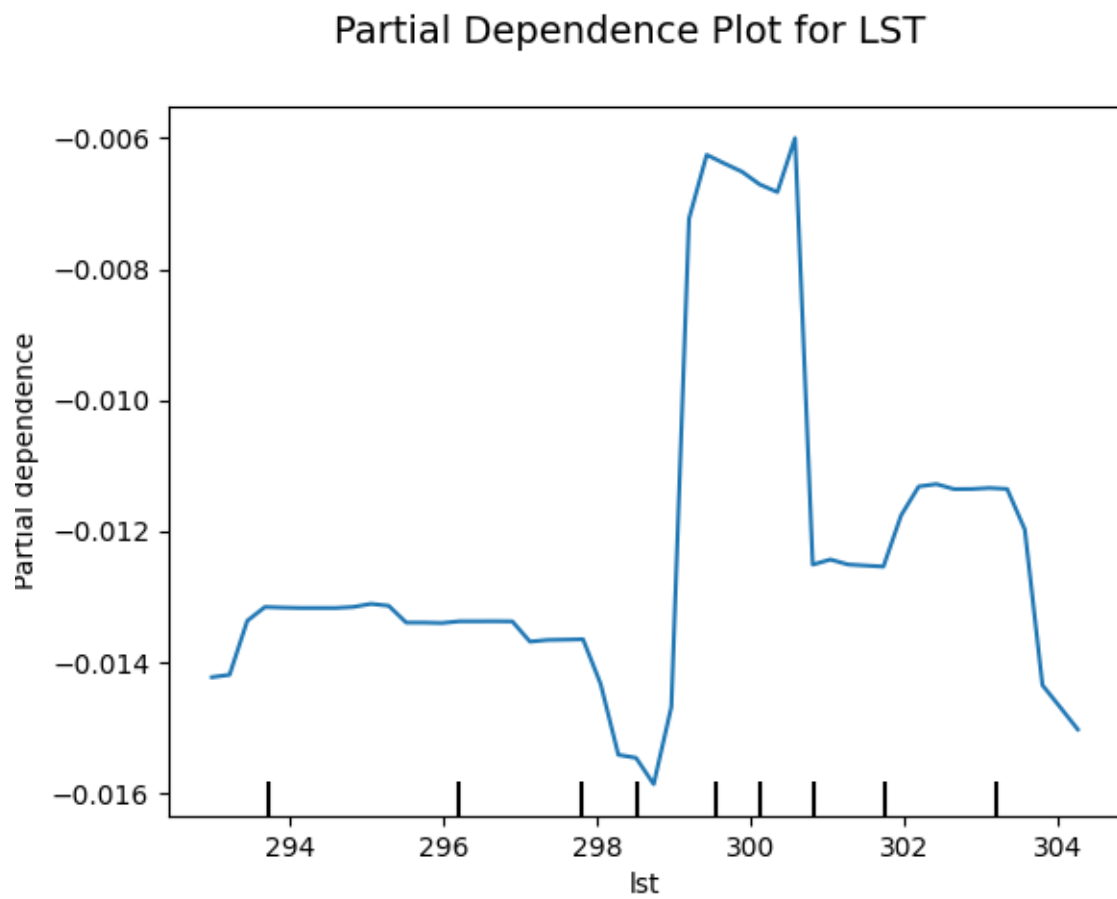


Figure 18: LST Partial Dependence Plot

4.11 Validation

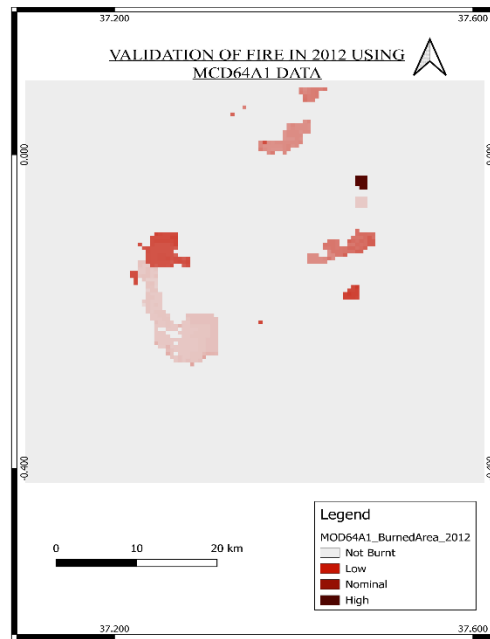
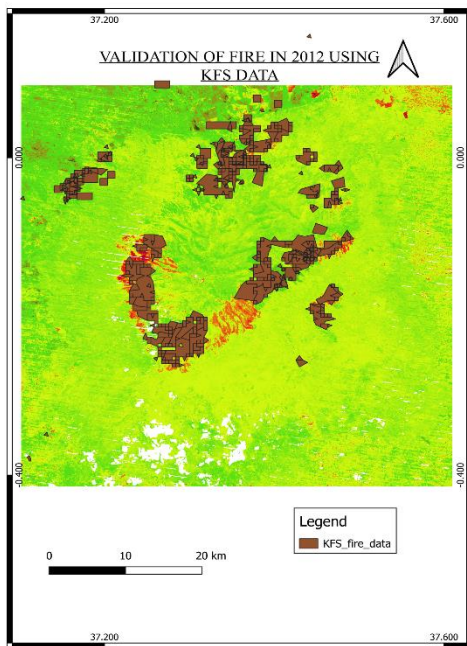


Figure 19: Burn Severity Map, 2012 Overlaid with KFS Data

Figure 20: Burn Severity Map, 2012 Overlaid with MCD64A1

CHAPTER 5: Discussions

5.1 Model Evaluation Metrics

5.1.1 RMSE and R^2

Root Mean Square Error (RMSE): The RMSE value of 0.028 indicates the level of error in the model's predictions. It is a commonly used metric to quantify the difference between predicted and observed values. A low RMSE value suggests that the model's predictions are relatively accurate, which is crucial when evaluating vegetation recovery following fires. The relatively small RMSE in this study reflects the model's ability to predict recovery dynamics with high precision.

R-Squared (R^2): An R^2 of 0.901 indicates that 90.1% of the variation in the post-fire vegetation recovery, as measured by NBR, can be explained by the model's predictors. This high R^2 value suggests that the model is highly effective in capturing the key factors influencing vegetation recovery. It demonstrates that the relationship between the variables used in the study (such as burn severity, temperature, and soil properties) accounts for most of the observed variance in the recovery process.

5.1.2 K-Fold Cross-Validation Results

The K-fold cross-validation procedure involved splitting the data into 10 subsets (folds) to assess the model's robustness and generalizability. The average RMSE across the 10 folds was found to be 0.0406, which is slightly higher than the RMSE from the test set (0.028), suggesting a small degree of overfitting. However, this RMSE is still considered to be within acceptable limits for this type of analysis, indicating that the model generalizes well across different data splits.

The R^2 value of 0.8753 across the folds further reinforces the model's predictive power. Despite the slight overfitting indicated by the RMSE, the high R^2 suggests that the model consistently captures the main relationships between the features and the target variable (post-fire recovery) across different subsets of data.

5.2 Statistical Analysis

5.2.1 ANOVA Results

Analysis of Variance (ANOVA) results showed a significant F-statistic of 12.16 with a p-value of $1.76e-10$, which strongly rejects the null hypothesis. This indicates that there are statistically significant

differences in vegetation recovery across the different burn severity classes. Burn severity, therefore, plays a critical role in determining the extent and rate of post-fire vegetation recovery. The significance of this finding aligns with previous studies that have shown a clear relationship between fire intensity and the recovery process, where more severe fires typically result in slower recovery rates.

5.2.2 Tukey HSD Test

The Tukey HSD (Honest Significant Difference) test revealed that pronounced differences were found between Group 1 (Unburned) and Groups 3, 4, 5, 6, and 7 (representing Moderate-Low to Very High Severity), highlighting that as burn severity increases, the recovery trajectory becomes more distinct, with severe burns leading to longer and more unpredictable recovery times.

These findings emphasize the importance of incorporating burn severity as a primary factor in recovery models, as different severities require tailored post-fire management strategies. Areas with very high burn severity may need more intensive restoration efforts to facilitate faster recovery.

5.3 Residual Plot Analysis

5.3.1 Model Fit and Bias

The residual plot comparing the predicted and actual NBR values shows a strong alignment along the 1:1 line, indicating that the model's predictions are quite accurate. The narrow spread around the line suggests low bias, with minimal overestimation or underestimation of recovery.

The model appears to be well-calibrated, and the residuals do not indicate any significant problems with model bias or error. However, some residuals may still indicate outliers and further investigation may be needed to refine the model and ensure its robustness for a wider range of fire events and recovery conditions.

5.4 Partial Dependence Plot for LST

5.4.1 Observations

The Partial Dependence Plot (PDP) illustrates how Land Surface Temperature (LST) influences the model's predictions for post-fire recovery, particularly the relationship between LST and the normalized

burn ratio (NBR), which is commonly used as an indicator of vegetation health.

The plot reveals that LST has a relatively stable, negative relationship with recovery at lower temperatures (below ~ 299 K), with a sharp decrease in recovery rates. However, at higher temperatures, especially above 299 K, the relationship becomes more complex:

- Around 300 K, there is a sharp increase in the partial dependence, indicating a possible positive influence of slightly warmer conditions on recovery. However, this is followed by high volatility and a general decline in recovery at temperatures exceeding 304 K, which highlights the negative impact of extreme heat on vegetation recovery.

5.4.2 Implications for Post-Fire Management

This analysis suggests that moderate LST conditions (around 299–300 K) are conducive to vegetation recovery, but higher temperatures above 300 K can lead to greater volatility and a negative impact on recovery. Management strategies should focus on mitigating extreme LST conditions, particularly in areas prone to very high temperatures due to factors such as global warming or altered land use patterns.

It is recommended to enhance soil moisture retention and reduce surface temperatures in severely burned areas by using techniques such as shading, mulching, and planting temperature-tolerant species. These strategies could help to stabilize the recovery process, especially in regions experiencing higher than average temperatures.

5.5 Post-Fire Vegetation Recovery Over the Years

5.5.1 Temporal Trends

The temporal analysis of vegetation recovery from 2012 to 2021 indicates a slow and steady recovery from the catastrophic fire in 2012. The NBR change over time shows:

- A sharp decline in 2012, with an average drop of -0.15, indicative of the significant damage caused by the fire. This dramatic decrease is expected, as fires often result in extensive vegetation loss, which would be captured by a negative change in NBR.
- Recovery began to stabilize from 2013 to 2016, as the NBR values started to level off near zero, suggesting that initial recovery processes were underway, albeit slow.

- The period from 2017 to 2021 shows positive recovery trends, with a notable peak in 2020, reflecting the cumulative effects of successful recovery mechanisms. This trend may also reflect the decreasing frequency of fire events and improved climatic conditions, leading to a more favorable environment for recovery.

5.5.2 Factors Influencing Recovery

The study identified that environmental factors such as temperature and precipitation likely played a role in the observed recovery patterns. Areas experiencing more favorable weather conditions in the years following the fire (such as increased rainfall) showed more significant recovery. Additionally, areas with higher elevations, where temperatures are generally lower, appeared to recover more rapidly compared to lower-elevation regions.

These findings suggest that vegetation recovery is not a uniform process but varies across space and time. Therefore, post-fire recovery strategies should consider not only burn severity but also temporal and spatial variations in environmental factors. Effective monitoring and adaptive management are essential for supporting recovery efforts.

5.6 Correlation Matrix Analysis

5.6.1 Correlation Findings

NBR change is positively correlated with mean temperature, max temperature, and LST, suggesting that warmer conditions are linked with higher recovery in certain burn severity classes.

Total carbon (TOTC) is negatively correlated with recovery, implying that soil carbon content may play a role in moderating vegetation regrowth post-fire.

Elevation is strongly correlated with temperature variables, indicating that higher elevations are associated with cooler temperatures, which likely benefit recovery processes.

5.6.2 Implications for Recovery Models

These correlations underscore the complex interactions between climate, soil, and topography in influencing post-fire recovery. While temperature and carbon content are critical factors, elevation also plays

a significant role in shaping recovery outcomes. Future models could benefit from considering these interactions more explicitly, perhaps through the inclusion of interaction terms or non-linear modeling approaches.

5.7 Validation

Data	Burnt Area
Landsat	23400ha
MCD64A1	22388ha
KFS	9878ha

Table 3: Validation of 2012 fire

CHAPTER 6: Conclusions

6.1 Summary of Key Findings

This study has provided comprehensive insights into post-fire vegetation recovery dynamics through advanced modelling and statistical analysis. The developed model demonstrated high predictive accuracy ($R^2 = 0.901$, RMSE = 0.028) in capturing vegetation recovery patterns, successfully integrating multiple environmental variables to explain recovery trajectories.

Several critical findings emerged from this research:

First, burn severity emerged as a fundamental determinant of vegetation recovery patterns. Statistical analysis revealed significant differences in recovery rates across burn severity classes ($p < 1.76e-10$), with more severely burned areas requiring longer recovery periods and showing more unpredictable recovery patterns. This finding emphasizes the need for severity-specific approaches in post-fire management strategies.

Second, the study revealed a nuanced relationship between Land Surface Temperature (LST) and vegetation recovery. Optimal recovery occurred within a narrow temperature window around 299-300K, with decreased recovery rates at both lower and higher temperatures. This temperature sensitivity highlights the potential vulnerability of post-fire recovery to climate change and extreme temperature events.

Third, temporal analysis over the 2012-2021 period demonstrated the long-term nature of vegetation recovery, with initial sharp declines followed by gradual improvement. The recovery trajectory showed distinct phases, from immediate post-fire decline through stabilization to eventual positive recovery trends, providing valuable insights into the temporal dynamics of ecosystem regeneration.

6.2 Management Implications

The strong influence of burn severity on recovery outcomes suggests that immediate post-fire assessment of burn severity should be a priority in developing management strategies. Areas with high burn severity may require more intensive intervention and monitoring to support recovery.

Temperature management emerges as a critical factor in recovery success. Management strategies should incorporate measures to moderate surface temperatures in recovering areas, particularly through techniques such as

mulching, strategic shading, and the selection of temperature-resilient species for replanting efforts.

The demonstrated importance of elevation and its relationship with temperature and recovery rates suggests that management approaches should be tailored to topographic conditions. Higher elevation areas may require different management strategies compared to lower elevation sites due to their distinct recovery patterns.

6.3 Future Recommendations

Enhancing Data Resolution: The use of higher-resolution satellite imagery would help improve model accuracy and reliability.

Exploring Temporal Variability: Future research could examine how the influence of variables like LST and soil organic carbon changes over time, particularly during the recovery period.

Additional research incorporating longer time series and climate change scenarios would enhance our understanding of future recovery dynamics.

Recovery Dynamics of Different Species: Future research should investigate species-specific recovery dynamics, as different plant species may have varying levels of resilience to fire depending on factors such as regeneration capacity, root depth, and moisture requirements. This could involve integrating species-level data into the recovery models to refine predictions and allow for more tailored conservation and restoration strategies.

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Appendix

